2.27 Numerical Methods: Change of Sign Method

Worked Solutions 1–10

1 a)

$$\mathrm{f}(x) = x^3 - 4x - 1$$

$$\mathrm{f}(2) = 8 - 8 - 1 = -1, \qquad \mathrm{f}(3) = 27 - 12 - 1 = 14$$

Since $\mathrm{f}(x)$ continuous and there is a change in sign, then there is a root.

1 b)

$$\mathrm{f}(x) = 2 + x - e^x$$

$$\mathrm{f}(1) = 2 + 1 - e^1 = 0.2817\ldots$$

$$\mathrm{f}(2) = 2 + 2 - e^2 = -3.389\ldots$$

Since $\mathrm{f}(x)$ continuous between $x = 1$ and $x = 2$ and there is a change in sign, then there is a root.

1 c)

$$\mathrm{f}(x) = x - 4 - 2\ln x$$

$$\mathrm{f}(8) = 8 - 4 - 2\ln 8 = -0.1588\ldots$$

$$\mathrm{f}(9) = 9 - 4 - 2\ln 9 = 0.6055\ldots$$

Since $\mathrm{f}(x)$ continuous between $x = 8$ and $x = 9$ and there is a change in sign, then there is a root.

1 d)

$$\mathrm{f}(x) = 1 - x \times 5^x$$

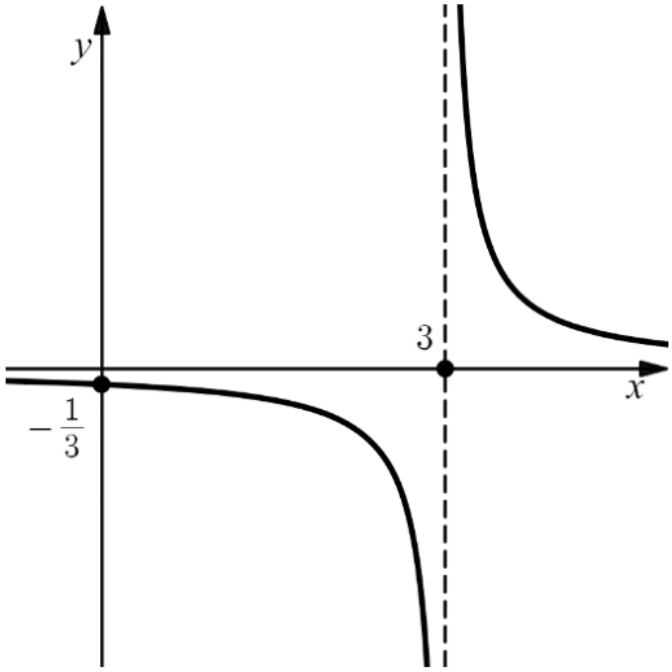
$$\mathrm{f}(0) = 1 - 0 \times 5^0 = 1, \qquad \mathrm{f}(1) = 1 - 1 \times 5^1 = -4$$

Since $\mathrm{f}(x)$ continuous and there is a change in sign, then there is a root.

2 a)

$\mathrm{f}(2) = \frac{1}{2-3} = -1$, $\mathrm{f}(4) = \frac{1}{4-3} = 1$

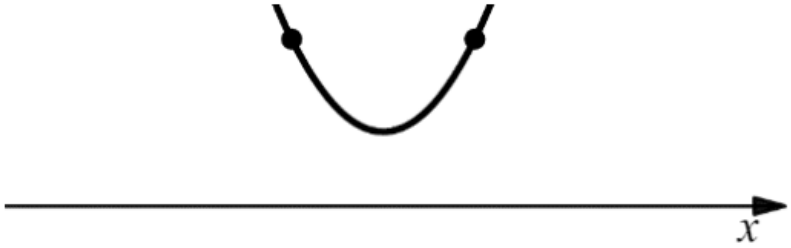
2 b)



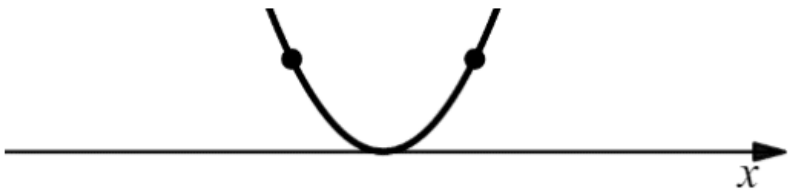
2 c)

Whilst there is a change in sign between the two values of x, the graph is not continuous for the interval, at $x = 3$ there is a vertical asymptote so we cannot determine if there is a root or not.

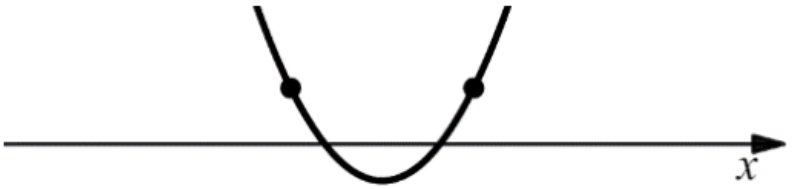
3 a)



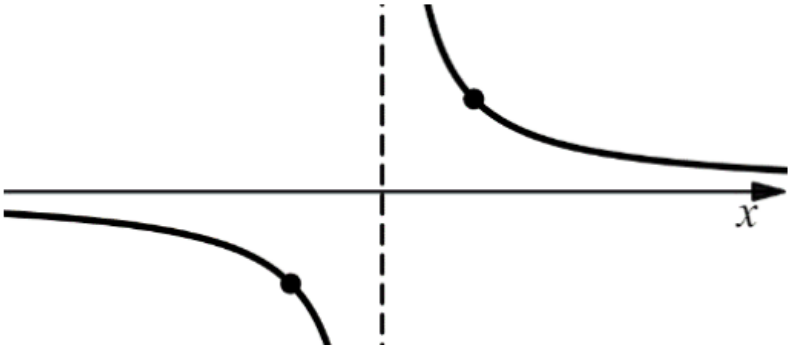
3 b)



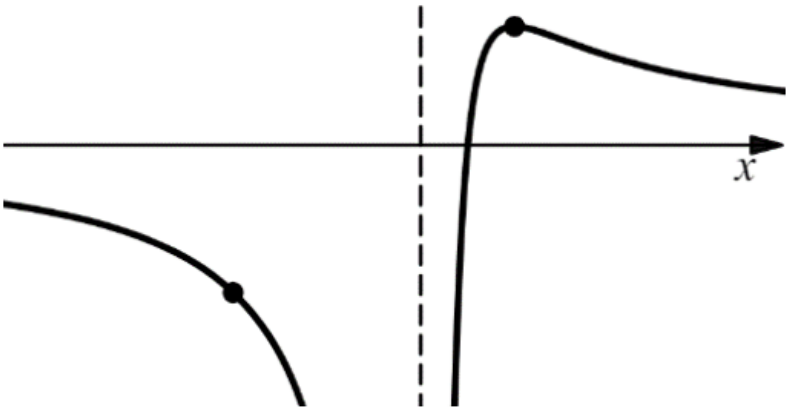
3 c)



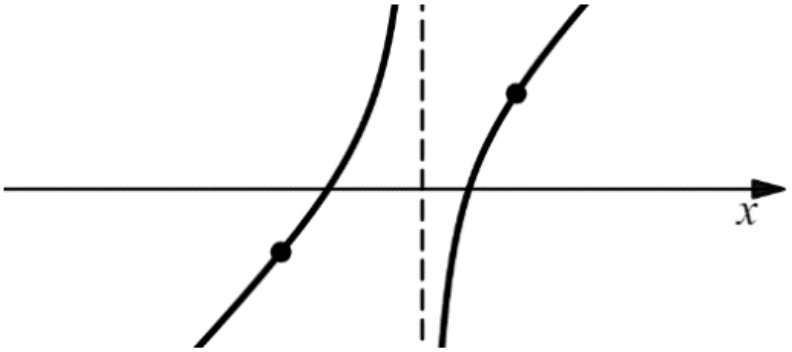
4 a)



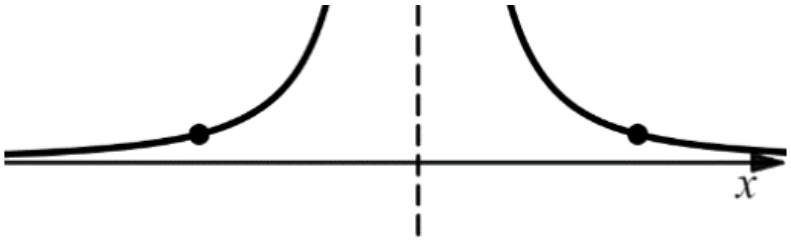
4 b)



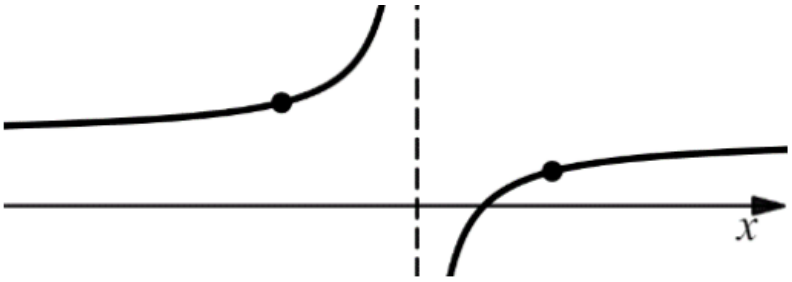
4 c)



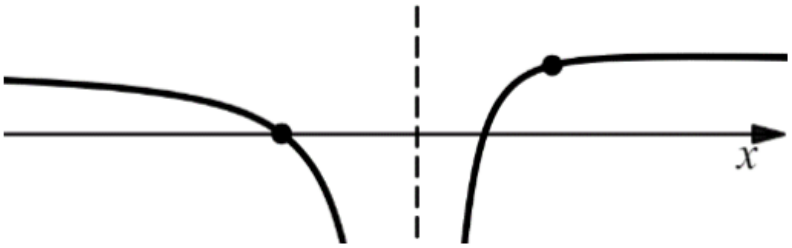
4 d)



4 e)



4 f)



5 a)

$$\mathrm{f}(-1) = -\tfrac{1}{2} - \tan(-1) = 1.0574\ldots$$
$$\mathrm{f}(1) = \tfrac{1}{2} - \tan(1) = -1.0574\ldots$$
$$\mathrm{f}(2) = \tfrac{1}{4} - \tan(2) = 2.4350\ldots$$

5 b)

There is an asymptote at $x = 0$ and $x = \frac{\pi}{2}$ so the curve is discontinuous between each of the pairs of points, therefore we cannot determine if there are any roots.

6 a)

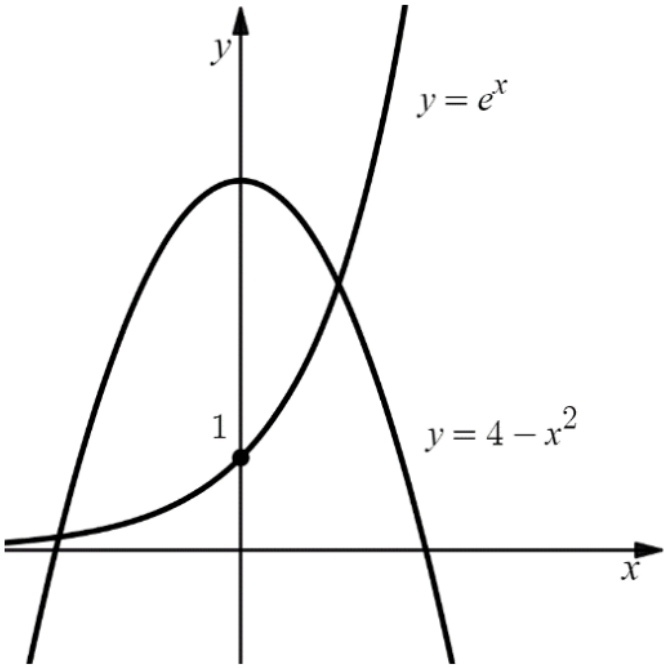
$$\mathrm{f}(x) = e^x - \frac{1}{x}$$
$$\mathrm{f}(0.5) = e^{0.5} - \frac{1}{0.5} = -0.3512\ldots$$
$$\mathrm{f}(1) = e^1 - \frac{1}{1} = 1.718\ldots$$

Since f(x) continuous between $x = 0.5$ and $x = 1$ and there is a change in sign, then there is a root.

6 b)

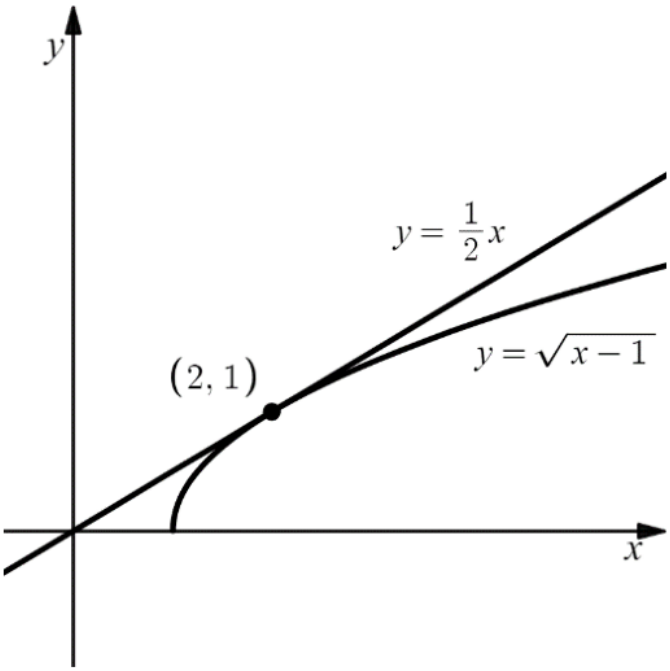
Whilst there is a change in sign between the two values of x, the graph is not continuous for the interval, at $x = 0$ there is a vertical asymptote so we cannot determine if there is a root or not.

7)



Two points of intersection ⇒ two solutions

8)



One point of intersection ⇒ one solution

9 a)

$$\frac{1}{x} = x^3 + 3 \Rightarrow 1 = x^4 + 3x \Rightarrow x^4 + 3x - 1 = 0$$

9 b)

$$\mathrm{f}(x) = x^4 + 3x - 1$$

$$\mathrm{f}(0) = 0^4 + 0 - 1 = -1, \qquad \mathrm{f}(1) = 1 + 3 - 1 = 3$$

Continuous between the two points, there is a change in sign therefore there is a solution between the two points.

9 c)

$$\mathrm{f}(x) = x^4 + 3x - 1$$

$$\mathrm{f}(-2) = (-2)^4 + 3(-2) - 1 = 9$$

$$\mathrm{f}(-1) = (-1)^4 + 3(-1) - 1 = -3$$

Continuous between the two points, there is a change in sign therefore there is a solution between the two points.

10 a)

$$x^2 = -\tfrac{1}{4}x^3 + 4 \Rightarrow \tfrac{1}{4}x^3 + x^2 - 4 = 0 \Rightarrow x^3 + 4x^2 - 16 = 0$$

10 b)

$$\mathrm{f}(x) = x^3 + 4x^2 - 16$$

$$\mathrm{f}(1.67855) = 1.67855^3 + 4 \times 1.67855^2 - 16 = -0.0005144\ldots$$

$$\mathrm{f}(1.67865) = 1.67865^3 + 4 \times 1.67865^2 - 16 = 0.001673\ldots$$

Continuous between the two points, there is a change in sign therefore there is a solution.

10 c)

1.6786